Independent component analyzis



zero shot meselasınfın yarısından örnek alındı diğer tarafta test edildi. diğer taraftan örnek alınmamış.

Metrikler

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}$$

Mesafe ölçümü ve metrikler:
Distance measure and metrics

Metrik:

① Negatif olmama:

$$d(x) \geq 0$$

② $d(x,y) = 0 \Rightarrow x = y$

③ $d(x,y) = d(y,x)$ simetri özelliği

④ $d(x,z) \leq d(x,y) + d(y,z)$

$\arg\max_x f(x)$, $f(x)$ max yapanı

$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}$ $d(x,y) = \sqrt{(x_1-y_1)^2 + (x_2-y_2)^2 + \cdots}$

Divergence: (Bu metrik değil ama o da mesafe ölçer) birisi diğerine daha uzakmı onu ölçüyor.

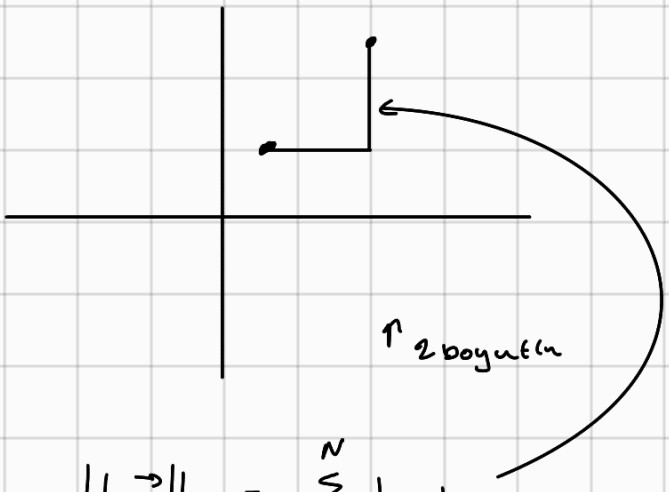
Tüm metrik özellikleri sağlamazlar

norm

$$\|\vec{x}\|_2 = \left(\sum_{i=1}^{N} |x_i|^2\right)^{1/2}$$

$$\|\vec{x}\|_p = \left(\sum_{i=1}^{N} |x_i|^p\right)^{1/p}$$

} ℓ_p-norm

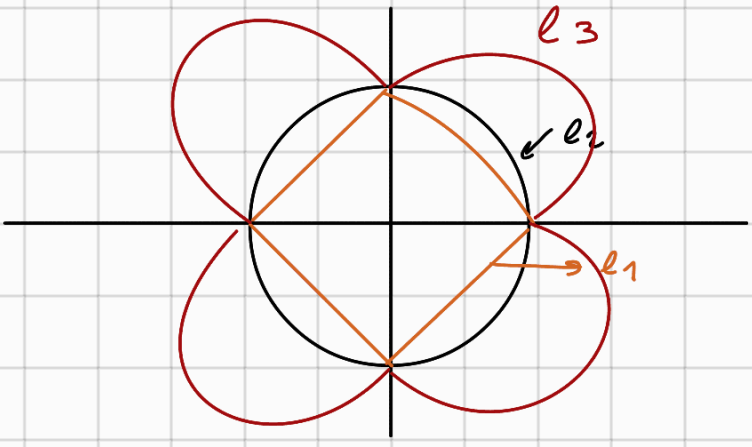


$$\|\vec{x}\|_1 = \sum_{i=1}^{N} |x_i|$$

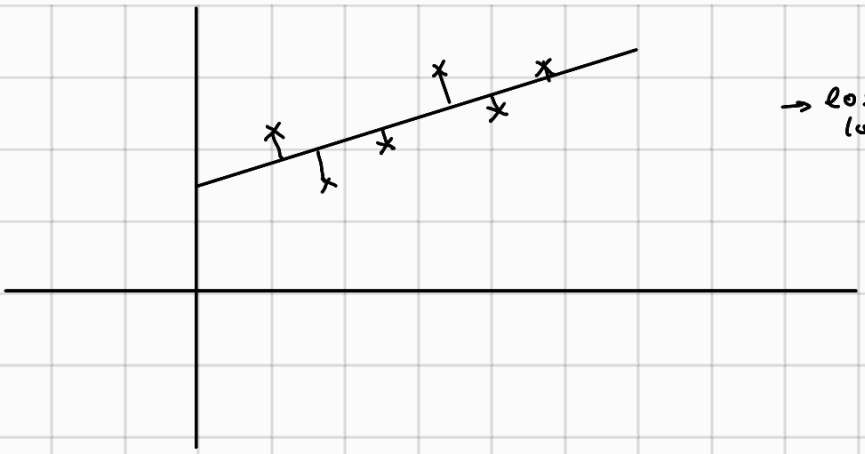
ℓ_1-norm: Manhattan norm

ℓ_2-norm: Öklid

ℓ_∞-norm: $\|x\|_\infty = \max_i |x_i|$



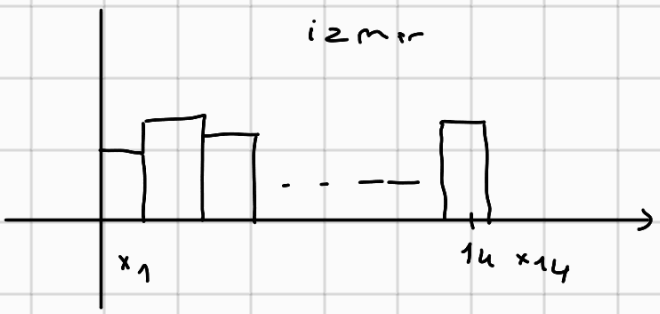
ℓ_p-normuna göre birim çemberler



→ loss function larda normlar kullanılıyor

$$\| \alpha x \|_p = |\alpha| \, \| x \|_p$$

Okuma alışkanlığını şehirlere göre

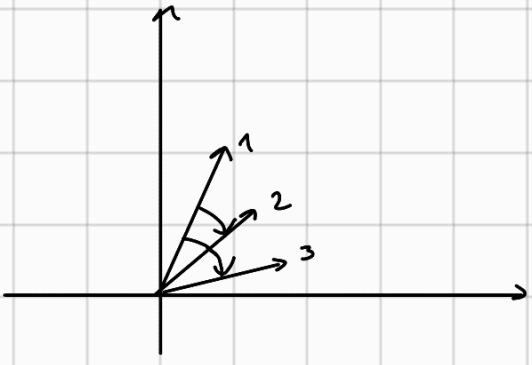


$$\vec{x}_{izmir} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{14} \end{bmatrix}$$



bardan sıkıntı na mesela nüfüsa bölebiliriz

$$\| \vec{x}_{izmir} - \vec{x}_{ankara} \|_2 > \| \vec{x}_{izmir} - \vec{x}_{antep} \|_2$$



$$\vec{x}_{izmir} \cdot \vec{x}_{ankara} = \frac{\| \vec{x}_{izmir} \| \, \| \vec{x}_{ankara} \| \cos\alpha}{\| \| \cdot \| \|}$$

Ama ben olasılık dağılımı üzerinden gitmek istiyorum

KL - Divergence:

$$D_{KL}(P \| Q) = \sum_{x \in X} P(x) \cdot \log \frac{P(x)}{Q(x)}$$

bir kez izmir ankara
bir kez izmir antep

Ama bu metrik değil

çünkü

1) negatif olabilir
2) simetrik değil $d(x,y) \neq d(y,x)$ her zaman

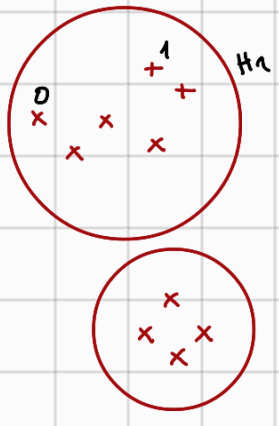
Metriğe çevirelim

$$D_{JS}(P,Q) = \begin{cases} \sum_{x \in X} P(x) \left| \log \frac{P(x)}{Q(x)} \right| \\ + \sum_{x \in X} P(x) \left| \log \frac{Q(x)}{P(x)} \right| \end{cases}$$

Jenson-Shannon div

$$M(x) = \frac{P(x) + Q(x)}{2}$$

$$\frac{1}{2} D_{KL}(P \| M) + \frac{1}{2} D_{KL}(Q \| M)$$



Entropi

$$H(x) = -\sum_{x \in X} P(x) \log P(x)$$

$$P(0) \cdot \log P(0) + P(1) \log P(1)$$

$$H_1 = -\left(\frac{2}{3} \log \frac{2}{3} + \frac{1}{3} \log \frac{1}{3}\right)$$

$$H_2(x) = 0 + 0 = 0$$

Nekadar çok kaos varsa entropi o kadar yüksek.

$P(0) = p \qquad P(1) = 1 - p \qquad p = 0,5$ max olur $H(x)$

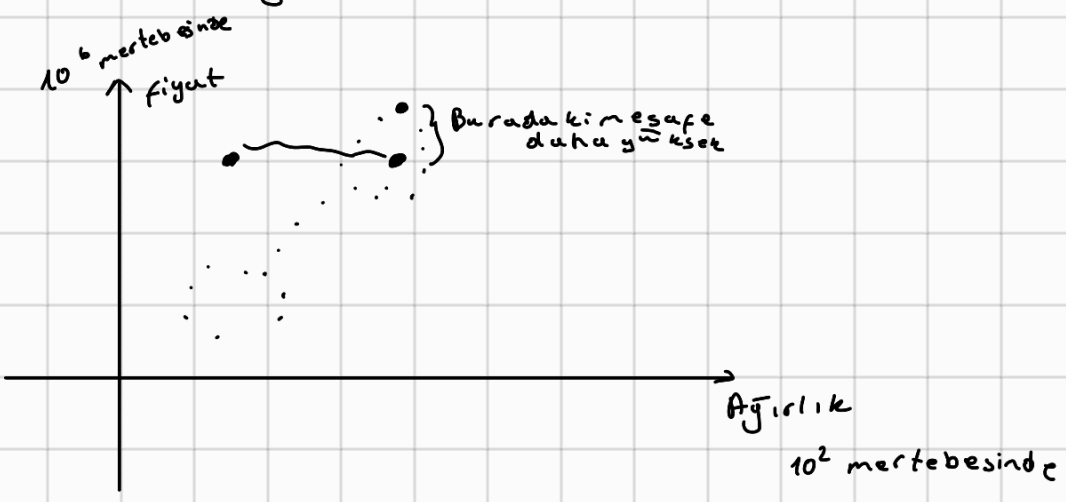
Continuous için:

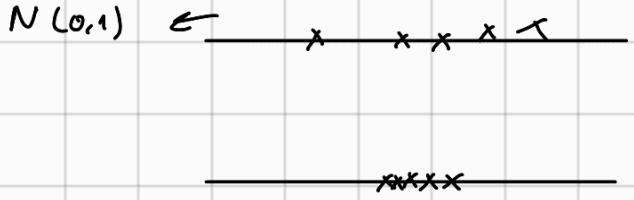
$$H(x) = -\int_{-\infty}^{+\infty} p(x) \log p(x)\, dx$$

Cross-Entropy:

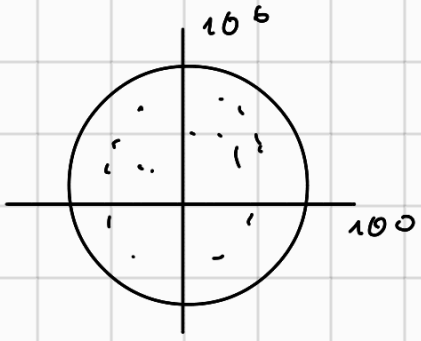
$$H(P, Q) = -\sum_{x \in X} P(x) \log Q(x)$$

P ile Q'nun birbirine uzaklığı





$$d_M(x, y) = \sqrt{(x-y)^T \underbrace{\Sigma_{xy}^{-1}}_{\text{kovariyans matrisinin tersi}} (x-y)}$$



$$\Sigma = \begin{bmatrix} 10^{12} & 0 \\ 0 & 10^4 \end{bmatrix}$$

Bir haber başkasına benziyor mu

iğdır'da trafik kazasından 10 kişi öldü

↓ Tokenization

iğdır trafik kaza 10 kişi ölmek

gs fb 3-1 yemek

iğdır avokado yetişmek

Küme Metrikleri

Hamming Distance

Bütün veriler tek (set())

$$d_H(A, B) = |A \cap B|$$

Jaccard distance:

$$1 - \frac{|A \cap B|}{|A \cup B|}$$

Sınıflandırma metrikleri

kestirim

etiket	0	1	
0	TN	FP	
1	FN	TP	

Doğruluk (Acc)

$$\frac{TN + TP}{\#\text{total}} = Acc$$

$$\{x^t, r^t\}_{t=1}^{N}$$

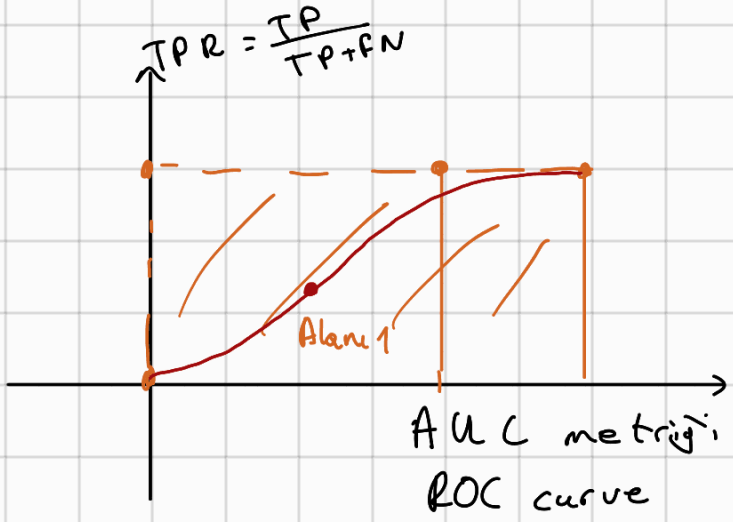
$$r^t = 0, \quad t = 1:N$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP+FN}$$

$$F\text{-}1\ score = 2 \times \frac{Prec \times Recall}{Prec + Recall}$$





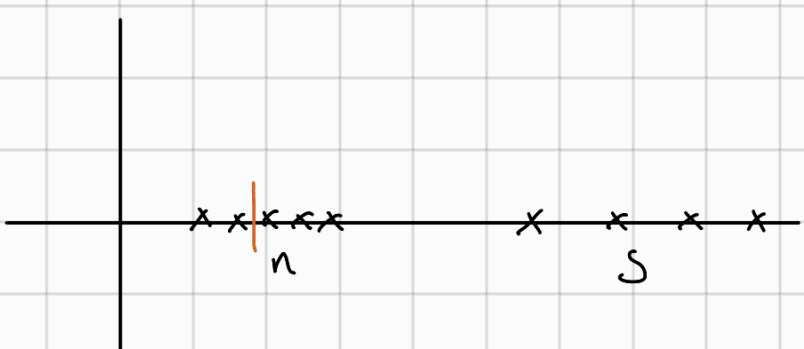
0'ı eşik değer yaparsam

0	5
0	4

$$TPR = \frac{4}{0+4} = 1$$

$$FPR = \frac{5}{5+0} = 1$$

$$FPR = \frac{FP}{TN+FP}$$



2	3
0	4

$$TPR = \frac{4}{4+0} = 1$$

$$FPR = \frac{3}{3+2} = \frac{3}{5}$$

Sora



5	0
0	4

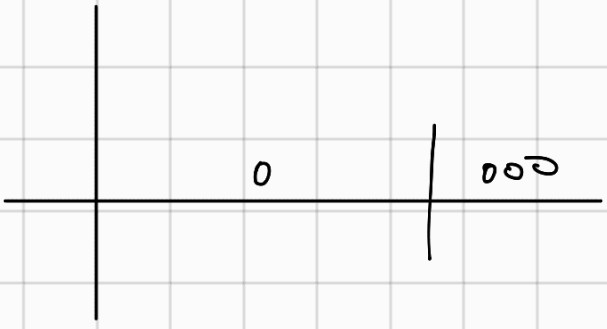
$TPR = \frac{4}{4+0} = 1$

$FPR = \frac{0}{0+5} = 0$

5	0
4	0

$TPR = 0$

$FPR = 0$



3	2
1	3

$TPR = \frac{2}{2+1} = \frac{2}{3}$

$FPR = \frac{2}{2+3} = \frac{2}{5}$

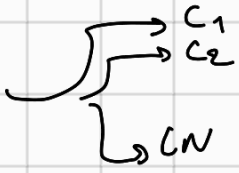
Confusion matrix

	a	b	c	d
a				
b				
c				
d				

$$\{x^t, r^t\}_{t=1}^{N}$$

$r^t \in \mathbb{C}$

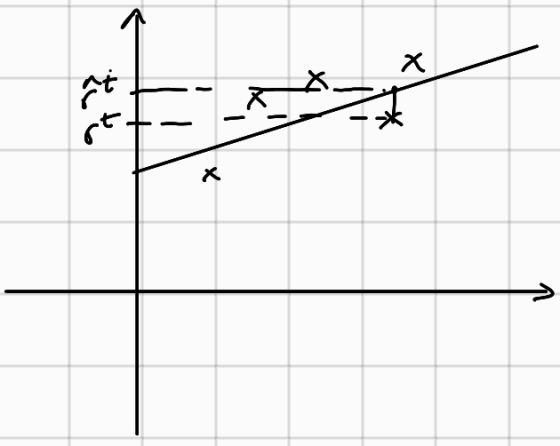
$r^t \in \mathbb{R}^N$



$f(x^t) = r^a$

$$\frac{1}{N} \sum_{t=1}^{N} (\hat{r}^t - r^t)^2$$

tahminler



$ax + b = f(x)$